

Research Paper

Stability and deformations of deposited layers in material extrusion additive manufacturing

Md Tusher Mollah^{*}, Raphaël Comminal, Marcin P. Serdeczny, David B. Pedersen, Jon Spangenberg

Department of Mechanical Engineering, Technical University of Denmark, Kgs. Lyngby, Denmark

ARTICLE INFO

Keywords:

Viscoplastic Materials
Material Extrusion Additive Manufacturing (MEX-AM)
Multiple-Layers Deposition
Computational Fluid Dynamics (CFD)
Deformation Control

ABSTRACT

This paper presents computational fluid dynamics simulations of the deposition flow during printing of multiple layers in material extrusion additive manufacturing. The developed model predicts the morphology of the deposited layers and captures the layer deformations during the printing of viscoplastic materials. The physics is governed by the continuity and momentum equations with the Bingham constitutive model, formulated as a generalized Newtonian fluid. The cross-sectional shapes of the deposited layers are predicted, and the deformation of layers is studied for different constitutive parameters of the material. It is shown that the deformation of layers is due to the hydrostatic pressure of the printed material, as well as the extrusion pressure during the extrusion. The simulations show that a higher yield stress results in prints with less deformations, while a higher plastic viscosity leads to larger deformations in the deposited layers. Moreover, the influence of the printing speed, extrusion speed, layer height, and nozzle diameter on the deformation of the printed layers is investigated. Finally, the model provides a conservative estimate of the required increase in yield stress that a viscoplastic material demands after deposition in order to support the hydrostatic and extrusion pressure of the subsequently printed layers.

1. Introduction

Three-dimensional printing of viscoplastic materials has grown in popularity over the recent years, due to the success of Material Extrusion Additive Manufacturing (MEX-AM) [1]. Viscoplastic materials, such as ceramic pastes [2,3], hydrogels [4], thermosets [5], and concrete [6], behave like solids when the applied load is below their yield stress, and like a fluid when the applied load exceeds their yield stress [7]. Viscoplastic materials are typically used in MEX-AM techniques such as Robocasting [8], and 3D concrete printing [9,10]. The differences between these technologies lie in the processing of the material before the extrusion and in the printing scale (from microscale to big area additive manufacturing). In these extrusion-based technologies, the structure is fabricated in a layer-by-layer approach onto a solid surface/support [11, 12]. During the process, the material is typically deposited on top of the previously printed layers that may be already solidified (wet-on-dry printing) or still deformable (wet-on-wet printing) [1].

In wet-on-wet printing, control over the deformation of layers is important for the stability and geometrical accuracy of the prints. If the

material is too liquid after the deposition, it cannot support the pressure of the subsequently deposited layers. On the other hand, the material flowability is a necessity during extrusion through the nozzle. Several experimental studies have been performed to analyze the physics of the extrusion and deposition of viscoplastic materials, as reviewed in Refs. [13–16]. The experimental measurements can be supplemented with Computational Fluid Dynamics (CFD) simulations to gain a more complete picture of MEX-AM. A review of the CFD studies within the material processing and deposition in 3D concrete printing was presented by Roussel et al. [17]. Wolfs et al. [18] predicted numerically the failure-deformation of a cylindrical structure due to the self-weight by calculating the stiffness and strength of the individual layers. It was found that the deformations can take place in all layers, however the most critical deformation occurs in the bottom layer. Comminal et al. [19,20] presented three-dimensional simulations of the material deposition in MEX-AM, where the fluid was approximated as Newtonian. Subsequently, the model was experimentally validated in Ref. [21] for polymer-based MEX-AM, and extended to simulate the deposition of multiple layers in Ref. [22], where the previously printed material was

^{*} Corresponding author.

E-mail addresses: mtumo@mek.dtu.dk, tusher.bsmrstu@gmail.com (M.T. Mollah).

<https://doi.org/10.1016/j.addma.2021.102193>

Received 22 December 2020; Received in revised form 1 May 2021; Accepted 15 July 2021

Available online 21 July 2021

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assumed solid. Xia et al. [23] simulated the influence of the viscoelastic effects on the shape of deposited layers in MEX-AM. A numerical model for simulating the deposition of a viscoplastic material was recently presented and experimentally validated in Refs. [24] and [25]. These studies focused on predicting the cross-sectional shape of a single printed layer for different processing conditions (relative printing speed, and layer height). Despite these research efforts, a limited number of studies have focused on investigating the material deformations in wet-on-wet printing when multiple layers are deposited on top of each other.

This paper presents CFD simulations of the extrusion-deposition flow of a viscoplastic material for several subsequent layers (viz. three- and five-layers). The material is continuously printed one layer over another on a fixed solid surface. The rheology of the viscoplastic material is approximated by the Bingham constitutive equation that is formulated using the Generalized Newtonian Fluid (GNF) model. The CFD model is used to predict the cross-sectional shapes of the layers and their deformations while printing the next layers on top. Moreover, the simulations are used to quantify the extrusion pressure applied by the deposited material on the substrate, and the previously printed layers. Numerically, it is investigated how the process parameters (i.e., the extrusion speed, printing speed, nozzle diameter, and layer height) and the material rheology affect the deformations of the deposited layers. Section 2 describes the methodology of the study. Section 3 presents and discusses the results. The study is summarized and concluded in Section 4.

2. Methodology

2.1. Governing equations and numerical model

The CFD simulation captures the extrusion-deposition flow of a viscoplastic material with the Bingham constitutive model. The fluid is extruded through a cylindrical nozzle with an inner diameter D , and it is deposited on a fixed solid surface. At time $t = 0$, the extrusion nozzle is at rest and located above the solid surface at a distance h , which is the theoretical layer height. At time $t > 0$, the material starts to flow through the nozzle with a fully developed profile and a constant average velocity U , called the extrusion speed. The nozzle remains at rest until the extruded fluid touches the substrate. Then, the extrusion nozzle starts to move with a constant velocity V , called the printing speed, along the horizontal x -direction. The material is extruded over a length l . After the layer is printed, the nozzle lifts up in the z -direction by the distance h , and the next layer is printed.

The model geometry along with the boundary conditions, nozzle toolpath, and the computational domain are presented in Fig. 1. The bottom plane of the domain is a stationary wall (the substrate). The top plane is an inlet boundary, where an artificial solid component obscures the most of the plane except the region that coincides with the nozzle

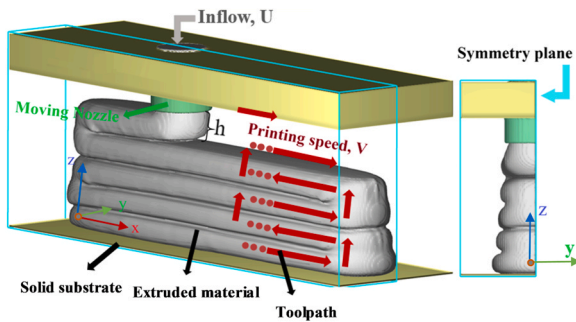


Fig. 1. Model geometry with the computational domain, extrusion nozzle, toolpath, and boundary conditions. The model is presented while printing the fifth layer.

orifice. A no-slip boundary condition is applied to all solid surfaces. The remaining planes have prescribed a continuative boundary condition that assumes a zero derivative of the velocity field across the boundary plane, and these planes are not in contact with the extruded material. Finally, the flow is assumed to be symmetrical with respect to the xz -plane.

The computational domain is meshed with a uniform Cartesian grid. The cell size is adapted in order to have between 25 to 32 cells along the nozzle diameter. The dimensions of the computational domain and the total number of discrete cells depend on the simulation case. The values of different processing parameters as well as meshing information for all the computational model cases are summarized in Table 1.

The flow is assumed to be transient and isothermal. The material is treated as incompressible with a constant density ρ . Therefore, the flow dynamics is governed by the continuity and momentum equations:

$$\nabla \cdot \mathbf{q} = 0 \quad (1)$$

$$\rho \left(\frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} \cdot \nabla \mathbf{q} \right) = -\nabla p + \rho \mathbf{g}_0 + \nabla \cdot \mathbf{S}_T \quad (2)$$

where $\mathbf{q}$ is the velocity vector, t is the time, p is the pressure, $\mathbf{g}_0$ denotes the acceleration vector due to the gravity $\mathbf{g}_0 = (0, 0, -g_0)$, and $\mathbf{S}_T$ represents the deviatoric stress-tensor.

The non-Newtonian flow behavior of viscoplastic materials can be predicted by a constitutive equation that accounts for the yield stress τ_0 , such as the Bingham [26], Casson [27], or Herschel-Bulkley (generalized Bingham) models [28]. In this study, the viscosity μ of the viscoplastic material is modeled with the Bingham rheological model:

$$\mu(\dot{\gamma}) = \begin{cases} \infty, & \text{for } \tau \leq \tau_0 \\ \eta_p + \frac{\tau_0}{\dot{\gamma}}, & \text{for } \tau > \tau_0 \end{cases} \quad (3)$$

where τ is the magnitude of the deviatoric stress-tensor, calculated as $\tau = \sqrt{2\text{tr}(\mathbf{S}_T^2)}$, η_p is the plastic viscosity, and $\dot{\gamma}$ is the strain rate. Eq. (3) prevents the flow of material when $\tau \leq \tau_0$.

The Bingham rheological model (see Eq. (3)) is used to calculate the deviatoric stress-tensor according to the generalized Newtonian fluid model:

$$\mathbf{S}_T = \mu(\dot{\gamma}) \mathbf{D}_T \quad (4)$$

where $\mathbf{D}_T = (1/2)((\nabla \mathbf{q}) + (\nabla \mathbf{q})')$ is the deformation rate tensor, the prime (') stands for the transpose notation, and $\dot{\gamma} = \sqrt{2\text{tr}(\mathbf{D}_T^2)}$. It should be noted that in this study, the rheological properties are assumed to be independent of time during the whole printing process.

The model represented by Eqs. (1) and (2) is solved numerically with the finite volume method, using the CFD software **FLOW-3D**, version 12.0 [29]. **FLOW-3D** has previously been successfully used for the simulations of the flow of viscoplastic materials [24,30]. The Bingham constitutive model (Eq. 3) is implemented using a bi-viscous regularization [31] (Eq. 5), to avoid the computational singularity, due to the infinite viscosity that appears in the theoretical model (Eq. 3). This is done by considering that the material have a maximum apparent viscosity $\mu_{\max,app}$ for shear rates $\dot{\gamma}$ smaller than a critical shear rate $\dot{\gamma}_c$. Thus, the apparent viscosity μ_{app} in the numerical model is:

$$\mu_{app}(\dot{\gamma}) = \begin{cases} \mu_{\max,app} = \eta_p + \frac{\tau_0}{\dot{\gamma}_c}, & \text{for } \dot{\gamma} < \dot{\gamma}_c \\ \eta_p + \frac{\tau_0}{\dot{\gamma}}, & \text{for } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \quad (5)$$

The viscous stress is solved implicitly by the successive under-relaxation technique, which allows to stabilize the convergence, when a high viscous stress gradient is observed. The pressure and velocity equations are solved implicitly in time to ensure the stability of the

Table 1

Processing and meshing parameters of the computational models for different simulation cases.

Model cases	Nozzle diameter D (mm)	Gap ratio h/D	l/D	Number of layers N	Printing speed V (mm/s)	Extrusion speed U (mm/s)	Speed ratio $S_r = V/U$	Computational domain dimensions (mm)	Number of cells	Cell size (mm)
Case 1	25	0.5	8	3	20–50	20–40	0.75–1.5	$250 \times 35 \times 88.5$	804,204	0.8
Case 2	50	0.5	8	3	20	20	1	$500 \times 70 \times 128$	780,624	1.78
Case 3	1.5	0.5	8	3	20	20	1	$15 \times 1.5 \times 5.35$	963,000	0.05
Case 4	0.4	0.5	12	3	20	20	1	$5.9 \times 0.6 \times 1.43$	1,473,920	0.015
Case 5	25	0.75	8	3	40	40	1	$250 \times 35 \times 95$	867,456	0.98
Case 6	25	1.0	8	3	40	40	1	$250 \times 30 \times 101$	760,530	1
Case 7	25	0.5	8	5	50	40	1.25	$250 \times 35 \times 105$	948,780	0.8

solution. The free surface tracking is captured by the Volume-Of-Fluid method (see e.g. [32,33] for details). An explicit solver calculated the momentum advection, which is second-order accurate in space and first-order accurate in time. The time step size is automatically controlled by the solver based on the stability criteria [29].

The material properties used in the simulations are presented in Table 2. Furthermore, The hydrostatic pressure applied by the printed layer to the bottom of the printed part is calculated as $P_H = N\rho g_0 h$, where N is the number of deposited layers and $g_0 = 9.81 \text{ ms}^{-2}$. Fig. 1 presents the simulated results of five successive layers using the parameters, $V = 50 \text{ mms}^{-1}$, $U = 40 \text{ mms}^{-1}$, $D = 25 \text{ mm}$, $h = 12.5 \text{ mm}$, $\tau_0 = 1000 \text{ Pa}$ and $\eta_p = 33 \text{ Pa.s}$. The material is continuously dispensed through a nozzle and creates a vertical 5-layers structure. It can be seen that the CFD model predicts the deformation of the printed layers while printing the next layers on top.

2.2. Post-processing

The CFD results are post-processed in MATLAB to obtain the cross-sectional shapes of the layers as well as the extrusion pressure. The cross-sectional shapes are used to monitor how the printed layers deform when other layers are printed on top, and the extrusion pressure is used to investigate the origin of the deformation. The cross-sectional shapes of the layers are reconstructed based on the fluid fractions in the computational cells. The fluid fractions are collected in a plane parallel to the yz -plane (see Fig. 1) that is located at the half of the length of the deposited layer ($l/2$) at the time step that corresponded to the end of each layer deposition. Next, the total relative width deformation of the bottom layer De_i after the deposition of the i -th layer is calculated as:

$$De_i = \frac{w_i - w_1}{w_1} \quad (6)$$

where w_i represents the width of the bottom layer just after the deposition of the i -th layer, and w_1 is the initial width of the bottom layer, as presented in Fig. 2.

The extrusion pressure P_E is the total pressure at the nozzle exit, which includes both the static and dynamic pressure. The extrusion pressure is applied by the extruded material to the closest surface (solid or fluid) [19]. In the case of the first layer, the nearest surface to the orifice is the substrate, while for the other layers, it is the already deposited material. To calculate P_E , the absolute pressure is extracted from the computational cells at the horizontal xy -plane passing through

Table 2

Material properties.

Parameter	Denoted by	Unit	Values
Density	ρ	kg.m^{-3}	1000
Dynamic yield stress	τ_0	Pa	100–2500
Plastic viscosity	η_p	Pa.s	12–900
Critical shear rate	$\dot{\gamma}_c$	s^{-1}	0.1

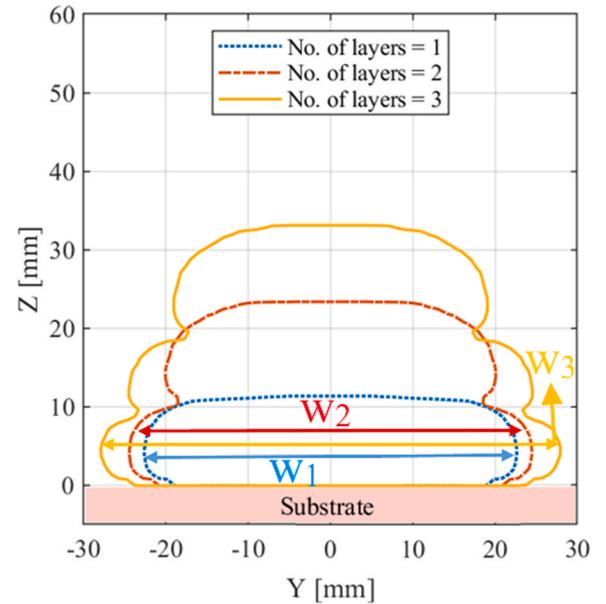


Fig. 2. Cross-sections after one, two and three deposited layers. w_1 , w_2 and w_3 are the width of the bottom layer after successive layer deposition.

the nozzle exit. A surface average of the absolute pressure is calculated at three time steps when the extrusion nozzle is approximately at the middle of the deposited layer. Next, the absolute pressure from those three time-steps are averaged and then subtracted by the atmospheric pressure (1 atm) to obtain the extrusion pressure.

3. Results and discussions

In this section, the influence of the constitutive parameters and process conditions on the cross-sectional shapes of the layers is investigated. Moreover, the deformation of the bottom layer is analyzed and related to the magnitude of the extrusion pressure.

3.1. Effect of the constitutive parameters

In Figs. 3–6, Case 1 (see Table 1) is simulated for different values of the yield stress and plastic viscosity. For reference, the hydrostatic pressure of a single deposited layer in the following simulations is $P_H = N\rho g_0 h \approx 123 \text{ Pa}$, where $h = 12.5 \text{ mm}$ and $N = 1$.

3.1.1. Effect of the yield stress

The cross-sectional shapes of the deposited layers are shown in Fig. 3 for varying yield stress. Also, the relative bottom layer deformation and the extrusion pressure are presented in Fig. 4 as a function of τ_0 , where $V = U = 40 \text{ mms}^{-1}$, and $\eta_p = 33 \text{ Pa.s}$. Fig. 3 shows that for the lowest yield stress ($\tau_0 = 100 \text{ Pa}$), a large deformation of layers is observed

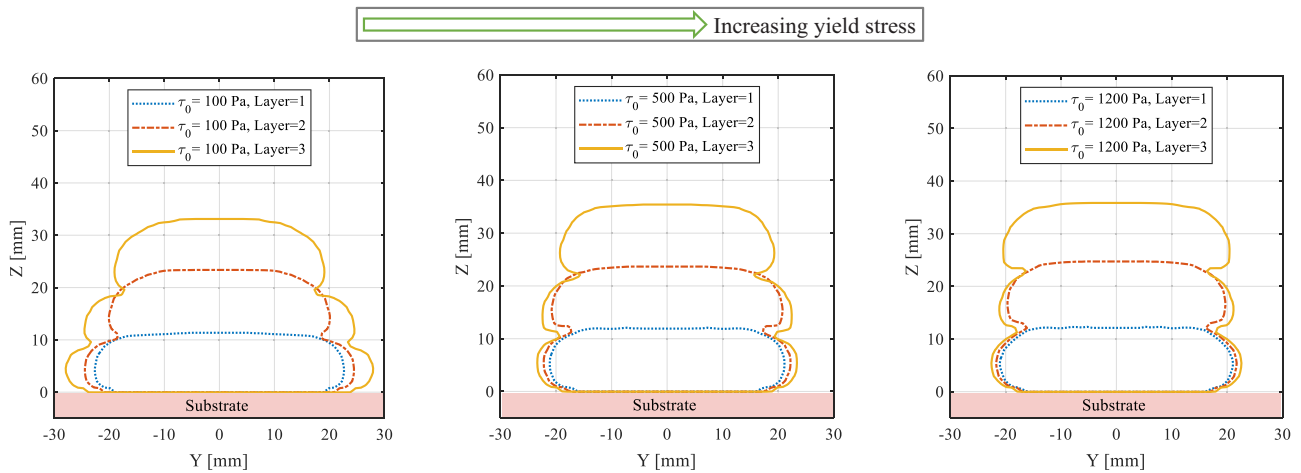


Fig. 3. Cross-sectional shapes for different yield stress (τ_0).

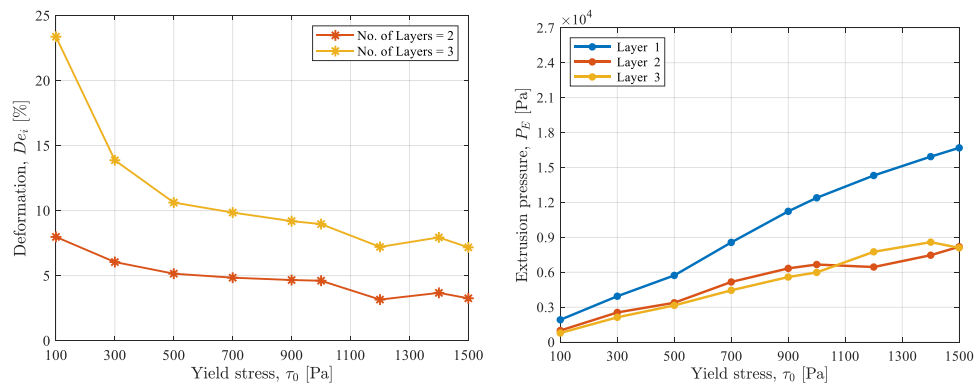


Fig. 4. Relative bottom layer deformation (De_l) (left) and extrusion pressure (P_E) (right) as a function of the yield stress (τ_0).

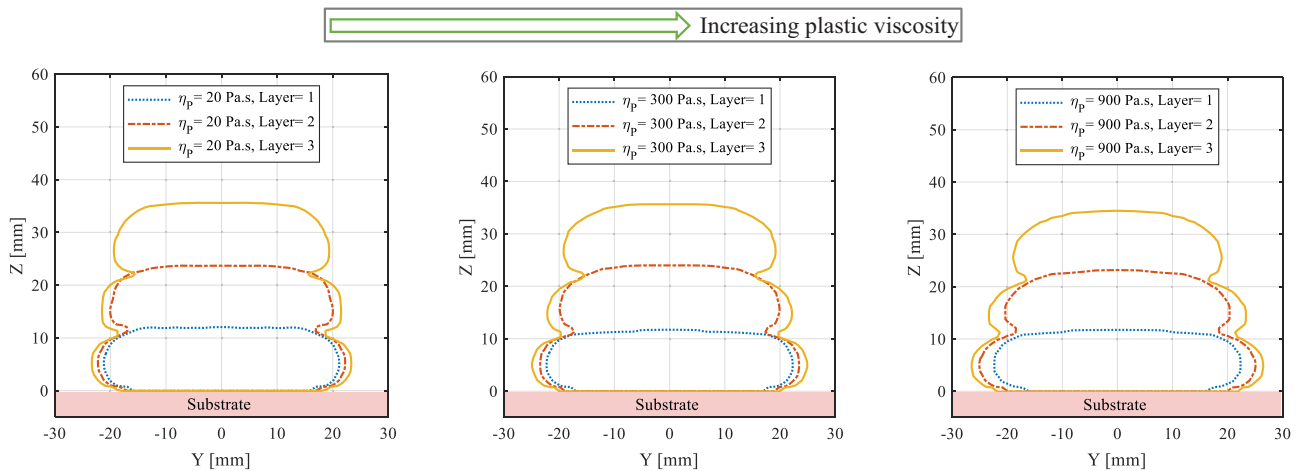


Fig. 5. Cross-sectional shapes for different plastic viscosity (η_p).

when the subsequent layers are deposited. This is due to the combined effect of the hydrostatic and extrusion pressure that exceed the yield stress of the material. When the yield stress is increased, less deformation of layers is observed, and the prints become more stable. This is due to the fact that a higher yield stress allows a higher load on the material before the deformation occurs.

Fig. 4 shows the relative bottom layer deformation (left) and the extrusion pressure (right) as a function of the yield stress. The relative deformation of the bottom layer is observed to decrease non-linearly

with the increasing yield stress. It is found that for the highest simulated yield stress, some deformations are still present. This is due to the extrusion pressure that increases with the yield stress, and becomes a driving force for the material deformations, next to the hydrostatic pressure. Note that the calculated extrusion pressure (Fig. 4(right)), is much larger than the hydrostatic pressure but it acts locally for a short period of time (as the nozzle moves), which limits its contribution to the total deformation. Fig. 4(right) also shows that P_E is higher when printing the first layer than the second and third layers. This is due to the

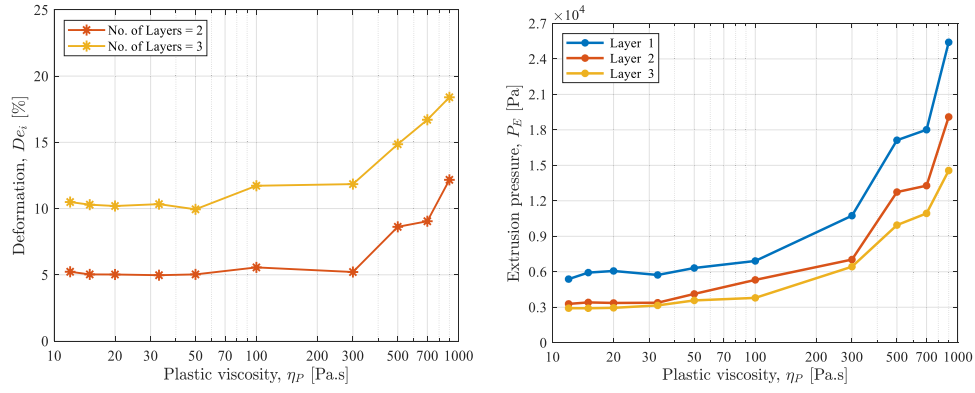


Fig. 6. Relative bottom layer deformation (De_i) (left) and extrusion pressure (P_E) (right) as a function of plastic viscosity (η_P).

fact that the first layer is deposited on a rigid substrate that does not deform whereas the second and third layers deform the previously deposited material, which reduces the pressure. In other words, the deformations of the sublayers increase the effective gap (i.e., the effective distance between the nozzle and previously printed material), which reduces the extrusion pressure required to spread the extruded material with shear deformations.

3.1.2. Effect of the plastic viscosity

The cross-sectional shapes, extrusion pressure and the relative bottom layer deformation for different values of the plastic viscosity are presented in Figs. 5 and 6. In these simulations $V = U = 40 \text{ mms}^{-1}$, and $\tau_0 = 500 \text{ Pa}$. In Fig. 5, it is seen that increasing the plastic viscosity leads to higher deformations in the printed layers. This can partly be attributed to the increase in extrusion pressure, see Fig. 6, which arises due to the higher resistance to flow (i.e., increased plastic viscosity) in the previous printed layers. Another contributing factor comes from the choice of applying a constant critical shear rate in all simulations, which will reduce the ratio between $\mu_{\max,app}$ and the average apparent viscosity during deposition, when increasing the plastic viscosity. This will lead to more uniform deformations throughout the layers. Fig. 6 also shows that the relative deformation of the bottom layer and extrusion pressure are almost independent of the plastic viscosity below 100 Pa.s, which indicates that the deformation at this point start to be dominated by the other processing parameters.

3.2. Effect of the processing parameters

In this subsection, the effect of the processing parameters (i.e., the

printing speed, extrusion speed, layer height, and nozzle diameter) on the cross-sectional shapes of the layers, the extrusion pressure, and the relative deformation of the bottom layer is discussed. Based on the results of the previous section, a large yield stress ($\tau_0 = 1000 \text{ Pa}$) and a small plastic viscosity ($\eta_P = 33 \text{ Pa.s}$) are chosen in order to pursue a stable printing scenario. Furthermore, the density was kept constant, as it was found to have a negligible effect on the extrusion pressure and deformation, as long as the hydrostatic pressure was below the yield stress.

3.2.1. Effect of the printing speed

The effect of the printing speed on the cross-sectional shapes for Case 1 (see Table 1), is presented in Fig. 7, where $U = 40 \text{ mms}^{-1}$. It can be observed that the layers become narrower, when the printing speed is increased, as there is less material deposited per unit length. Moreover, the layers deform less with the increase of the printing speed, as seen specifically for the bottom layer in Fig. 8(left). The reduced relative deformation is not a consequence of the hydrostatic pressure ($P_H = N\rho g_0 h$), as it does not change with the printing speed, but rather due to the fact that the extrusion pressure decreases with increasing print speed (see Fig. 8(right)) and it acts for a shorter time $t \sim D/V$. The extrusion pressure decreases because less material experience shear deformation per unit length.

3.2.2. Effect of the extrusion speed

The effect of the extrusion speed is presented in Figs. 9 and 10 for Case 1 (see Table 1), where $V = 30 \text{ mms}^{-1}$. The cross-sectional shapes that are presented in Fig. 9 become wider with an increasing extrusion speed, as there is more material deposited per unit length for a constant

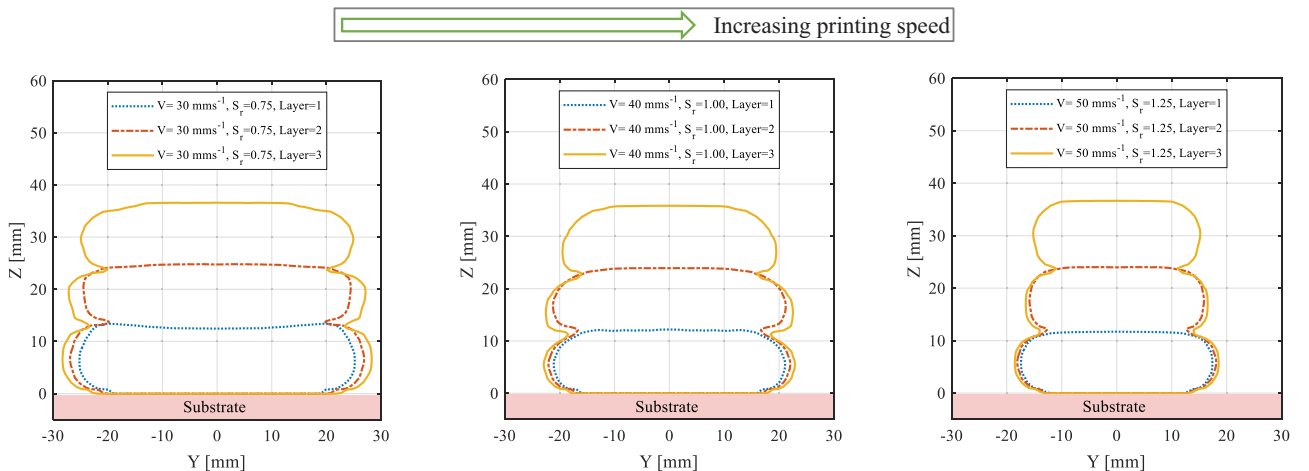


Fig. 7. Cross-sectional shapes for different printing speed (V).

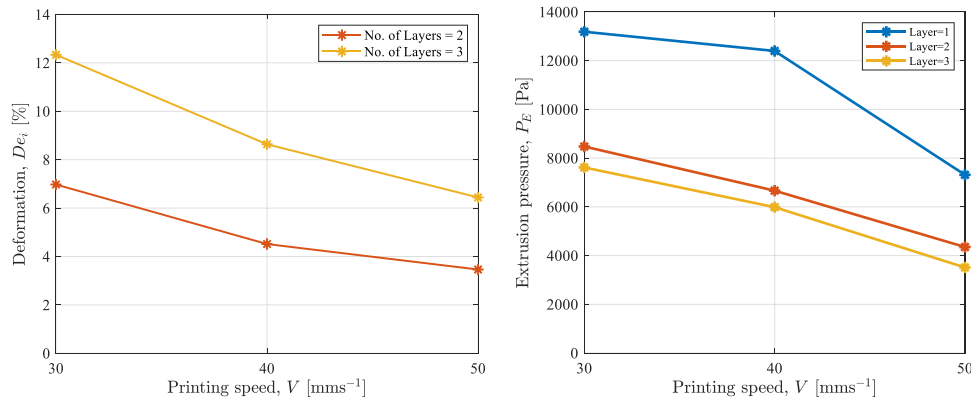


Fig. 8. Relative bottom layer deformation (De_i) (left) and extrusion pressure (P_E) (right) as a function of the printing speed (V).

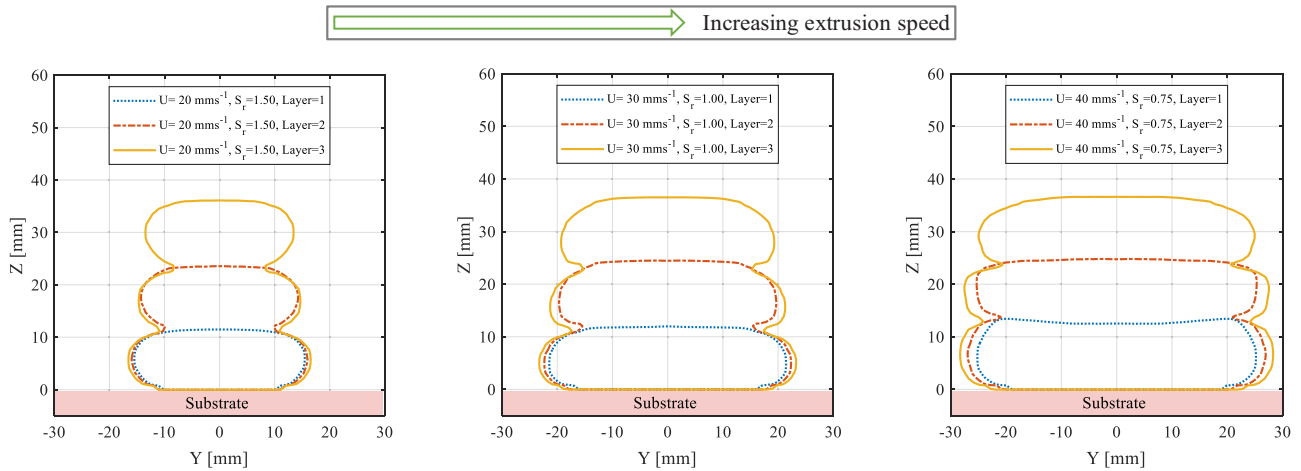


Fig. 9. Cross-sectional shapes for different extrusion speed (U).

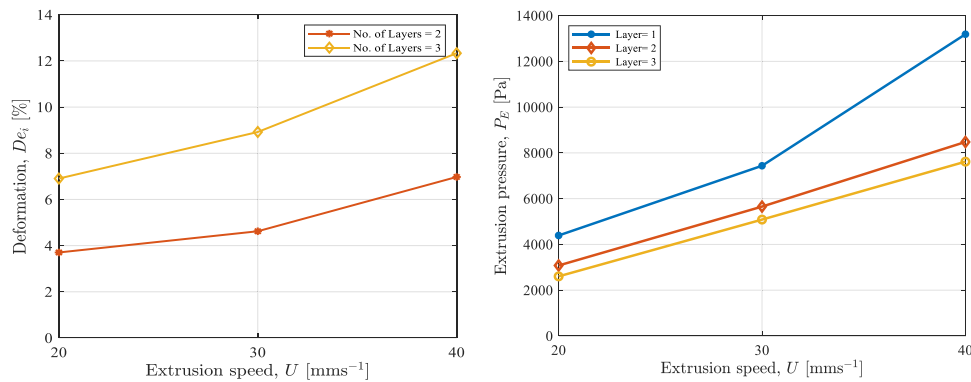


Fig. 10. Relative bottom layer deformation (De_i) (left) and extrusion pressure (P_E) (right) as a function of extrusion speed (U).

printing speed. Moreover, a larger deformation of layers is observed when increasing the extrusion speed. Fig. 10 shows the relative bottom layer deformation (left) and the extrusion pressure (right) as a function of the extrusion speed. As the hydrostatic pressure ($P_H = N\rho g_0 h$) does not change with the variation of the extrusion speed, the cause of the increasing deformation (Fig. 10(left)) is partly the increase in extrusion pressure, see Fig. 10(right), which is due to more material experiencing shear deformation per unit length. Another effect is that the increased extrusion speed leads to higher shear rates and thus reduce viscosities, which also increase deformations. The reduction in viscosities should also reduce the extrusion pressure, but that effect seems not to be

dominant in this case.

In the results presented in Figs. 7–10, it is seen that increasing the printing speed and the extrusion speed have opposite effects on the deformation of the layers, when the other parameters are held constant. Figs. 11 and 12 show the effect of increasing both the extrusion speed and printing speed at the same time such that the speed ratio is unchanged, $S_r = 1$. The opposing effects of these two parameters leads to a situation, where the relative bottom layer deformation decreases between $U = V = 20$ mms⁻¹ and $U = V = 30$ mms⁻¹, while the extrusion pressure is approximately constant; whereas for U and V between 30 mms⁻¹ and 40 mms⁻¹, De_i is approximately constant and P_E increases.

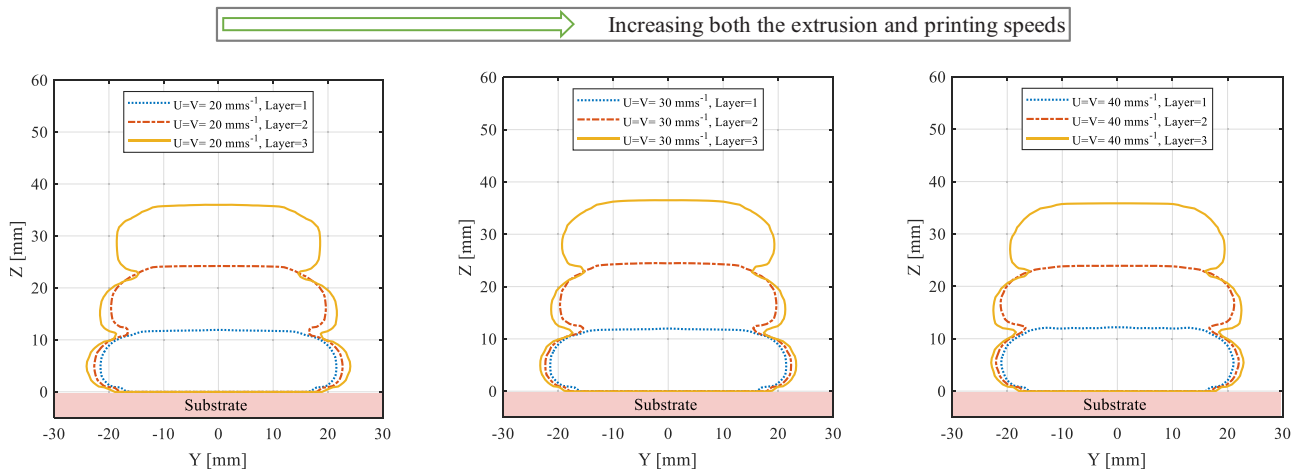


Fig. 11. Cross-sectional shapes for different pair of extrusion speed (U) and printing speed (V) with same speed ratio ($S_r = 1$).

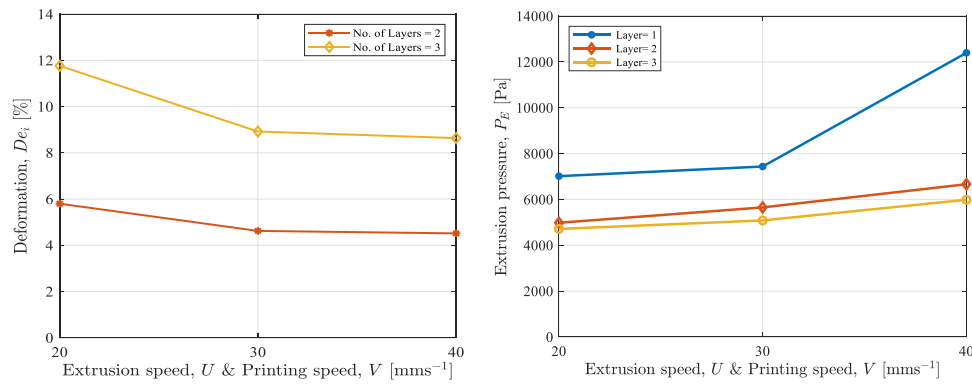


Fig. 12. Relative bottom layer deformation (De_i) (left) and extrusion pressure (P_e) (right) for different pair of extrusion speed (U) and printing speed (V) with the same speed ratio ($S_r = 1$).

Thus, within the investigated parameters range, the effect of increasing the printing speed has a dominating effect on the bottom layer deformation, while the increased extrusion speed dominates the extrusion pressure.

3.2.3. Effect of the layer height

In this subsection, the deposition of material is simulated for different layer heights, $h = 12.5$ mm (Case 1), 18.75 mm (Case 5) and

25 mm (Case 6), cf. Table 1, with $V = U = 40$ mm/s. Fig. 13 shows that the cross-sectional shapes of the layers become more rounded and narrower when the layer height is increased. Fig. 14 presents the relative bottom layer deformation (left) and the extrusion pressure (right) as a function of the layer height. The relative deformation De_i is observed to increase with the layer height, whereas the extrusion pressure is observed to decrease. The increase in De_i is however caused by the decreasing width of the bottom layer, when the layer height is increased.

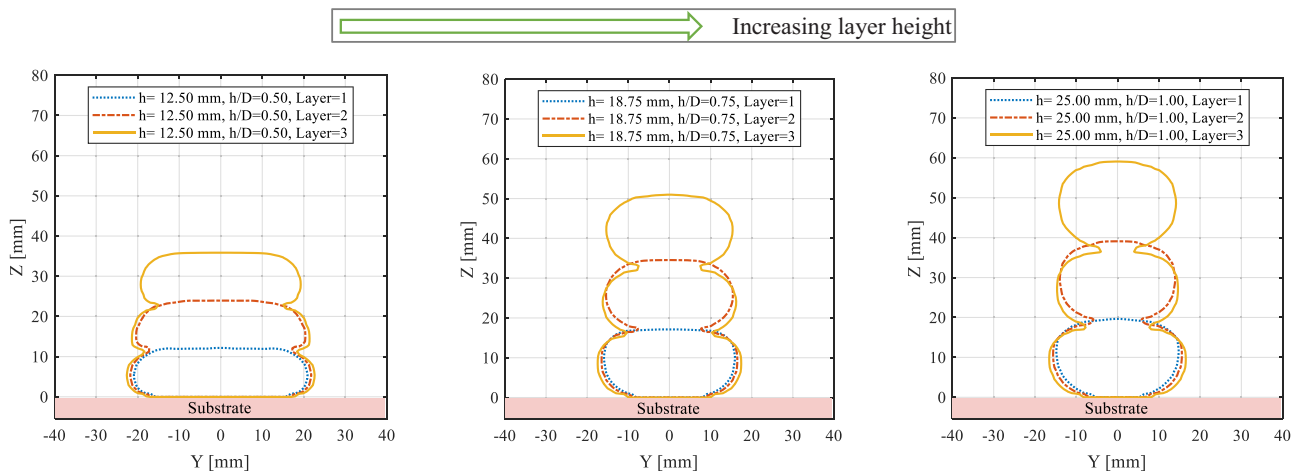


Fig. 13. Cross-sectional shapes for different layer height (h).

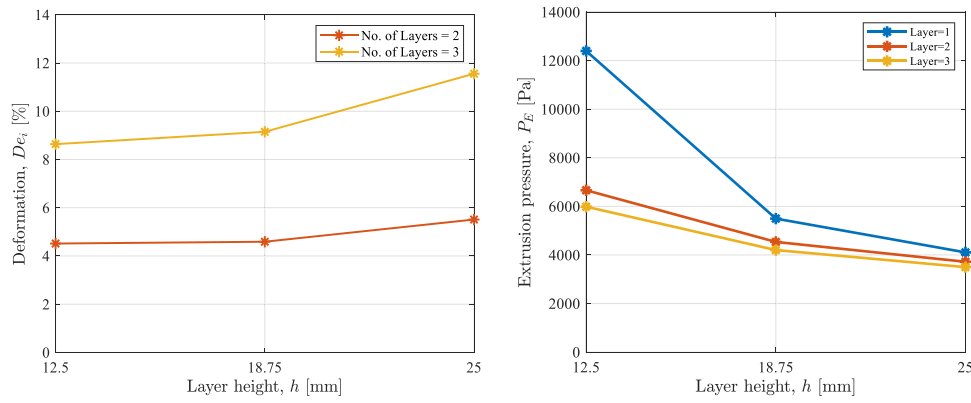


Fig. 14. Relative bottom layer deformation (D_{ei}) (left) and extrusion pressure (P_E) (right) as a function of layer height (h).

In fact, the absolute deformation of the bottom layer were similar for the different layer heights. For reference, the maximum hydrostatic pressure for three printed layers where $h = 25$ mm and $N = 3$ is $P_H \approx 736$ Pa, which is below the yield stress (1000 Pa).

3.2.4. Effect of the nozzle diameter

The influence of the nozzle diameter on the cross-sectional shapes of the deposited layers is presented in Fig. 15. The following nozzle diameters are investigated: $D = 0.4$ mm (Case 4), $D = 1.5$ mm (Case 3), $D = 25$ mm (Case 1), and $D = 50$ mm (Case 2), with the gap ratio $h/D = 0.5$ in each cases; see Table 1. The speeds $V = U = 20$ mm s⁻¹ are

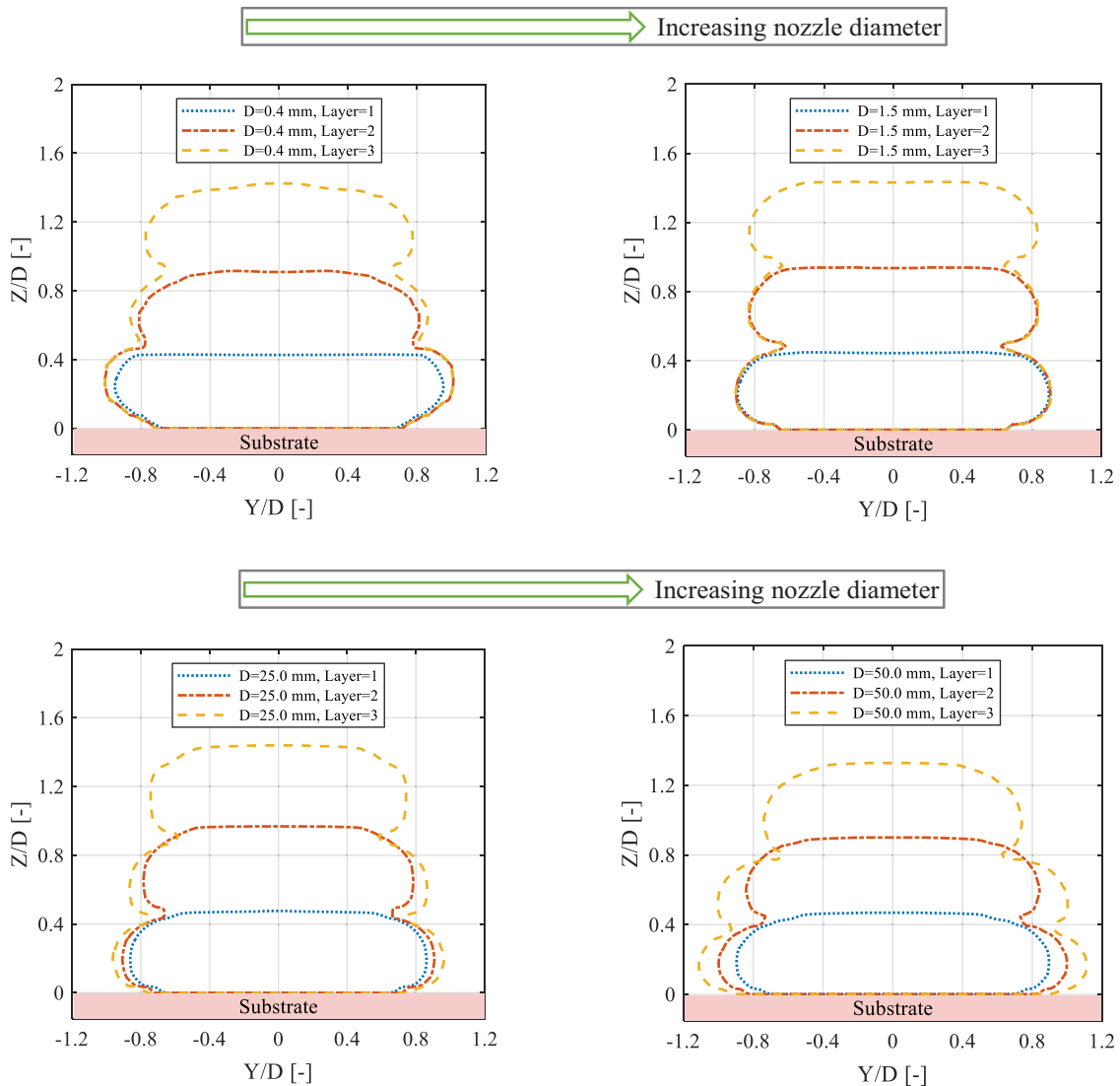


Fig. 15. Cross-sectional shapes for different nozzle diameter (D).

chosen to fit the range of processes represented by the different nozzle diameters. The axes in Fig. 15 are non-dimensionalized with the corresponding nozzle diameter to ease the comparison between the cross-sections. The results illustrate that the prints performed with larger diameters show larger deformation. The relative bottom layer deformation and extrusion pressure are shown in Fig. 16. The relative bottom layer deformation presents a minimum when $D = 1.5$ mm. This non-monotonic trend is due to the combined influence of the hydrostatic pressure and the extrusion pressure. The hydrostatic pressure ($P_H = N\rho g_0 h$, where $h = 0.5D$) increases with the nozzle diameter, while the extrusion pressure decreases with the increasing nozzle diameter, due to smaller shear stresses in the channels with larger cross-sectional area (see Fig. 16(right)). Moreover, as the printing speed is the same in the simulated cases, the extrusion pressure acts locally for a longer period of time, when the nozzle diameter is larger, which additionally contributes to the deformation.

3.2.5. Comparison of the effect of the processing parameters

In order to compare the effect of the parameters discussed above, a bar diagram is presented in Fig. 17. The amount of deformation of the bottom layer is shown as a function of different combinations of processing parameters. The combinations of the parameters are labeled with the symbol and magnitude of the parameters (excluding units), and refers to the simulations presented in Figs. 7–16. Fig. 17 shows that within the investigated parameter intervals, changes in the nozzle diameter have the largest effect on the deformation. Furthermore, the results illustrate that variations in layer height, extrusion speed, and printing speed leads to very similar sized deformations.

3.3. 5-Layers deposition

In this subsection, the CFD model is used to simulate the wet-on-wet deposition of 5 subsequent layers (Case 7 in Table 1) to investigate whether the same conclusions hold as for the presented 3-layers simulations. Based on the above results, the processing parameters: $D = 25$ mm, $h = 12.5$ mm, $h/D = 0.5$, $V = 50$ mm s⁻¹, $U = 40$ mm s⁻¹ and $S_r = 1.25$, are chosen in order to get an optimal print with less deformation. The results are presented for varying yield stress and a constant plastic viscosity, $\eta_p = 33$ Pa.s.

Fig. 18(left) presents the theoretical height ($N \times h$) and the actual height (simulated) of the part, as a function of the number of layers N , for different yield stress, $\tau_0 = 100, 500, 1000, 1800, 2500$ Pa. It is found that, in all cases, the actual height of the deposited material is lower than the theoretical height. Therefore, the effective distance between the nozzle exit and the previously deposited material (in simulations) is larger than the nominal layer height $h = 12.5$ mm. This effect is amplified as the number of layers increases. As a result, the cross-sectional shapes of layers become narrower and more circular with the deposition of each subsequent layer. This effect is also observed in

real-world prints [34]. To avoid the round and narrow shape of the new layers, the effective distance at each layer needs to be adjusted by the distance the previous layers have deformed. Fig. 18(right) presents the cross-sectional shapes of a 5-layer structure for uncorrected h (i.e. the effective distance increases after each layer), and for the corrected h (i.e. the nozzle height is adjusted after each layer to have the same effective distance h of 12.5 mm). It can be seen that the round and narrow shape of a new layer due to an increase in effective distance is mitigated by the corrected h . Thus, the following simulations are carried out with the corrected h .

The cross-sectional shapes for the different τ_0 are presented in Fig. 19, and show similar trends as for 3-layer deposition (see Fig. 3). The material is observed to continue deforming when two additional layers are added on top of the three-layer print. The relative bottom layer deformation is presented in Fig. 20(left) as a function of the layer number. It can be seen that increasing the yield stress can reduce the amount of deformation, but it never vanishes. This is because the higher yield stress produces a higher extrusion pressure that contributes to the deformation, see Fig. 20(right). Interestingly, for $\tau_0 = 1800$ Pa and $\tau_0 = 2500$ Pa, almost the same amount of layer deformations are predicted by the model, see Fig. 20(left). Thus, it is anticipated that a further increase of the yield stress would not result in significant improvements.

3.4. Wet-on-dry deposition of 5-layers

Within the range of the investigated parameters, the previous section showed that the deformation of layers cannot be completely eliminated, if the material has a yield stress that is constant in time. In order to eradicate the deformations, the deposited material should ideally increase its yield stress before the next layers are printed. In this section, the numerical model is used to study a situation where the previously printed material solidifies right after being deposited (i.e. wet-on-dry printing). This means that the already printed layers are regarded as solids in the computational model.

In Figs. 21 and 22, the cross-sectional shapes and extrusion pressure are shown, respectively, for simulations with 5 subsequent layers (i.e., Case 7 in Table 1). The simulations are carried out with three different yield stress values and $\eta_p = 33$ Pa.s. The cross-sectional shapes of the layers during the wet-on-dry printing are very stable, see Fig. 21. In the previous sections, it was observed that the extrusion pressure is the largest for the first layer (see Fig. 20), as it cannot deform the solid substrate. Fig. 22 supports this observation and shows that if the previously deposited layers do not deform, the extrusion pressure becomes the same for all the layers. The extrusion pressure results also provide important information about how much the materials' yield stress needs to increase in order to produce stable prints. The sum of the hydrostatic and extrusion pressure is a conservative estimate of the yield stress that is required by the previously deposited layer in order not to deform

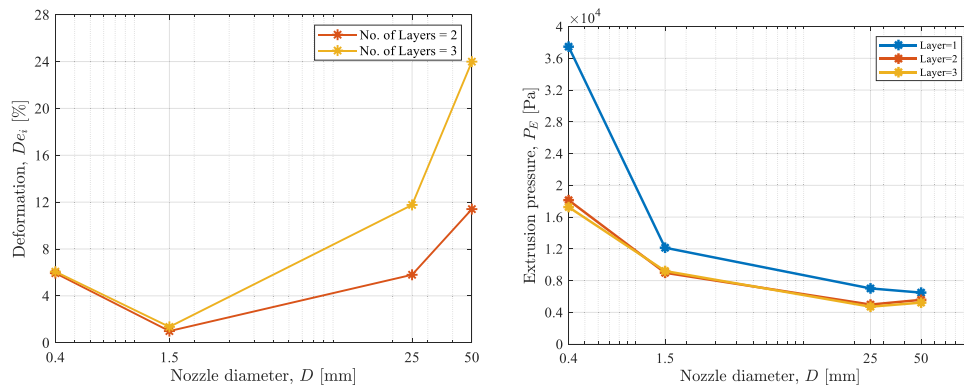


Fig. 16. Relative bottom layer deformation (D_{e_l}) (left) and extrusion pressure (P_E) (right) as a function of nozzle diameter (D).

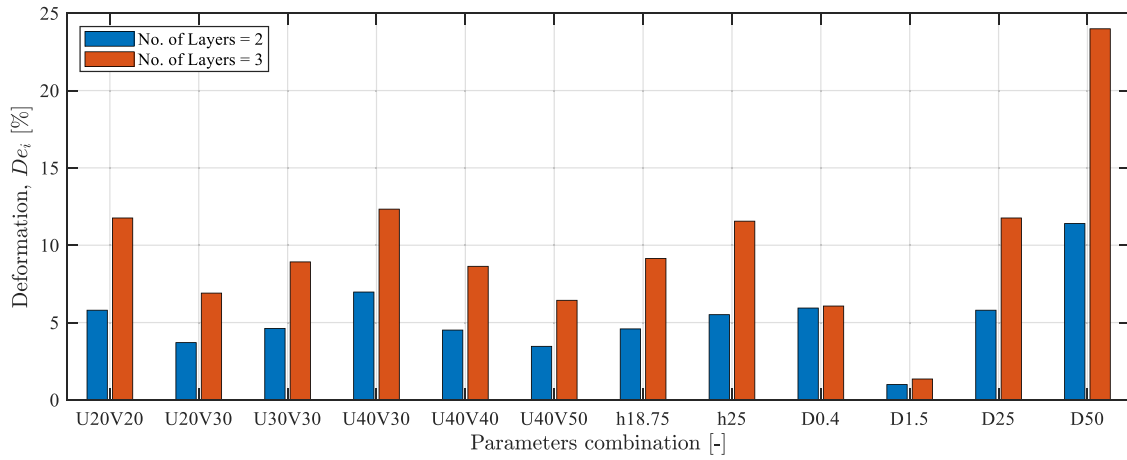


Fig. 17. Comparison among the parameters in terms of observed deformation.

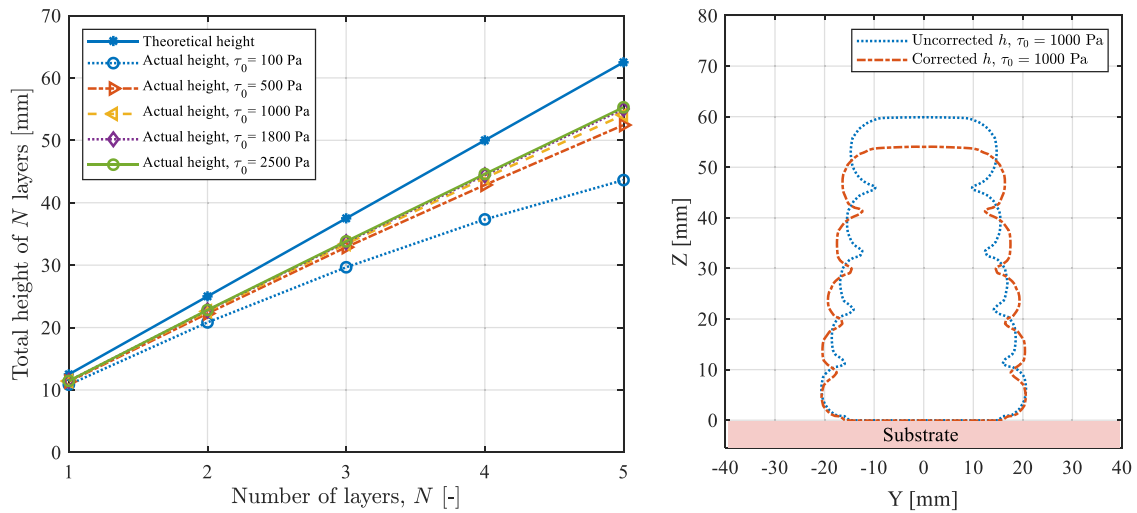


Fig. 18. Total height of N deposited layers (left) with corrected layer height, and cross-sectional shapes (right) for uncorrected and corrected layer heights in wet-on-wet deposition.

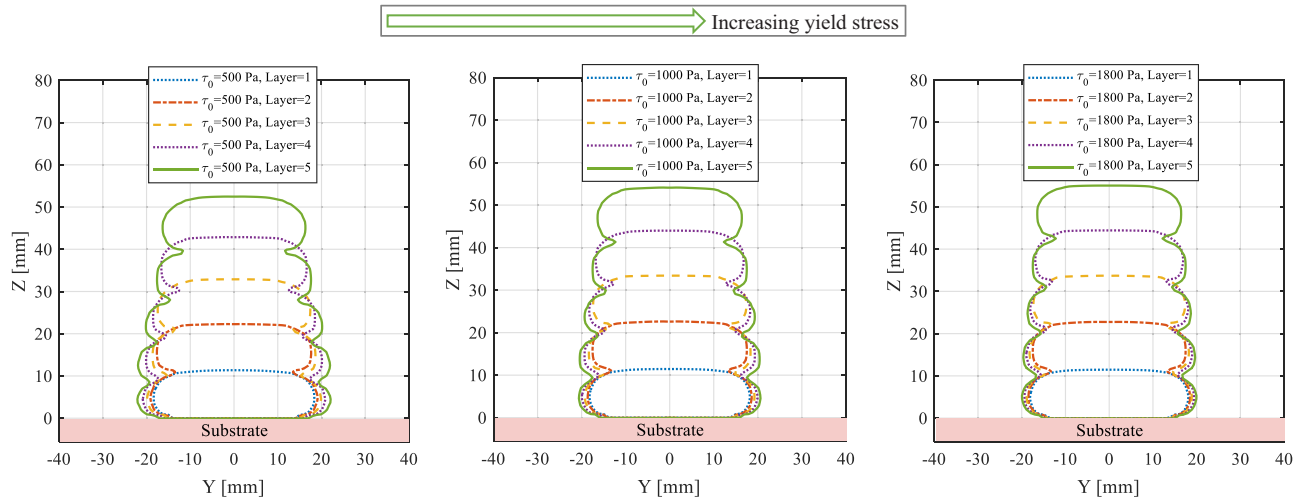


Fig. 19. Cross-sectional shapes for different yield stress (τ_0) in wet-on-wet deposition.

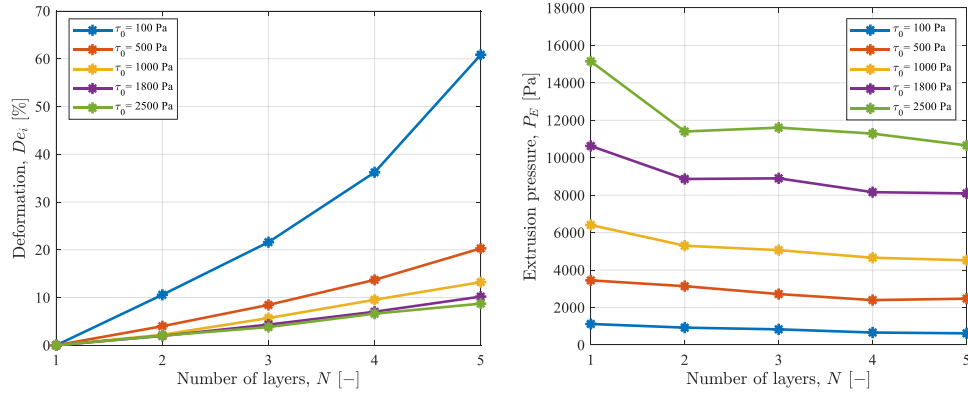


Fig. 20. Relative bottom layer deformation (D_{e_l}) (left) and extrusion pressure (P_E) (right) as a function of number of layers (N) in wet-on-wet deposition.

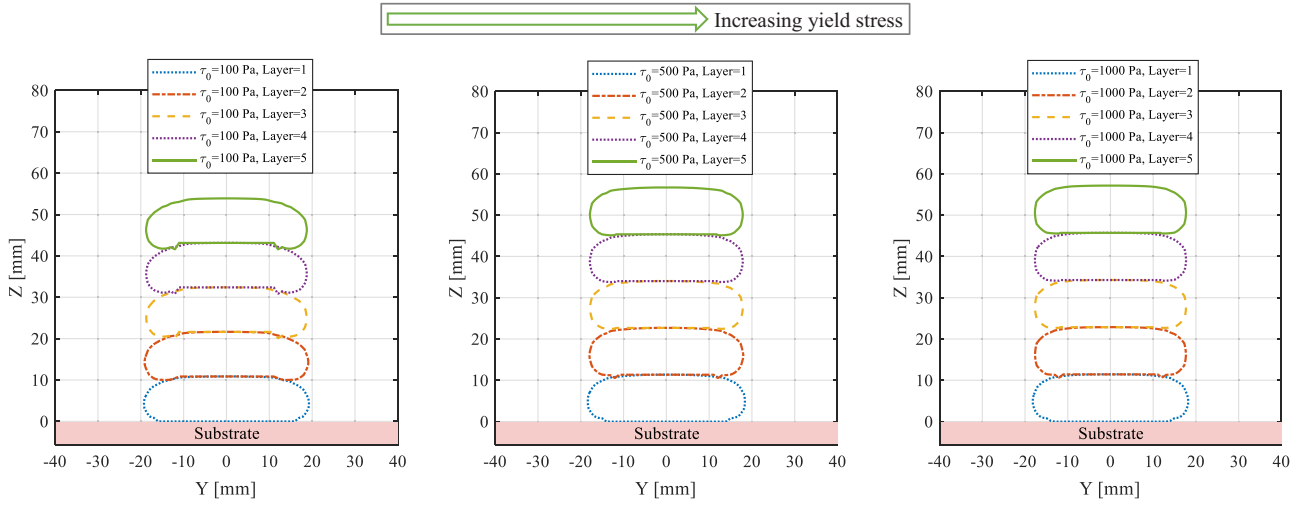


Fig. 21. Cross-sectional shapes for different yield stress (τ_0) in wet-on-dry deposition.

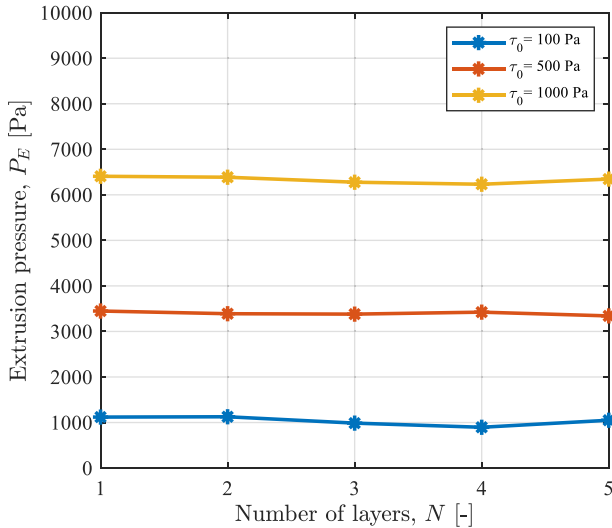


Fig. 22. Extrusion pressure (P_E) as a function of number of layers (N) in wet-on-dry deposition.

when subsequent layers are printed on top.

4. Conclusions

In this work, a CFD model was used to capture the extrusion-deposition flow of viscoplastic fluids during MEX-AM. The material was modeled with the Bingham constitutive equation, and the free surface of the fluid was tracked to quantify the deformation of layers during and after the deposition. The influence of the constitutive parameters and process conditions on the cross-sectional shapes of the layers were analyzed. It was found that when the yield stress is small, the layers deform due to the hydrostatic pressure generated by the subsequently deposited material, as well as the extrusion pressure of the extruded material. Increasing the yield stress of the material led to smaller deformations of the deposited layers. On the other hand, a higher plastic viscosity led to larger deformations of the layers, due to an increase in the extrusion pressure, as the material had a higher resistance to flow.

Increasing the printing speed resulted in smaller deformations of the layers, due to the decreasing extrusion pressure that acted for a shorter time, whereas increasing the extrusion speed behaved oppositely with regards to deformations and extrusion pressure. When the printing speed and extrusion speed were increased at the same time, a combination of the two opposing effects was observed for the investigated range of parameters. Furthermore, the increase of the layer height led to larger relative deformation of the layers, but similar absolute deformations. The dependency of the layer deformation on the nozzle

diameter was non-monotonic, due to the combined effect of the hydrostatic and extrusion pressure. When the nozzle diameter was increased, the hydrostatic pressure increased as the layer height increased with the nozzle diameter, while the extrusion pressure decreased due to the smaller shear stresses in larger channels. In all of the investigated cases of the wet-on-wet deposition, some deformations of the bottom layer were observed.

Finally, a wet-on-dry deposition of 5-layers was simulated, where the previously deposited material was assumed to be solidified before the deposition of the next layer. In that case, the extrusion pressure is the same for all layers and it was possible to get a stable print. Furthermore, from the wet-on-dry simulations, one could extract a conservative estimate for the required yield stress increment that the material needs to develop before the deposition of a subsequent layer, in order not to deform. The yield stress of the previous layers must be higher than the sum of the hydrostatic and extrusion pressure. In future work, the presented CFD model could be used to predict the deposition of materials, where the rheological behavior changes continuously as a function of time, which could lead to a less conservative yield stress criterion.

CRediT authorship contribution statement

Md. Tusher Mollah: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing. **Raphaël Comminal:** Conceptualization, Formal analysis, Writing – review & editing, Supervision. **Marcin P. Serdeczny:** Writing – review & editing, Formal analysis. **David B. Pedersen:** Supervision, Resources, Funding acquisition. **Jon Spangenberg:** Conceptualization, Formal analysis, Writing – review & editing, Supervision, Resources, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to acknowledge the support of the Danish Council for Independent Research (DFF) | Technology and Production Sciences (FTP) (Contract no. 8022-00042B). Moreover, the authors thank **FLOW-3D** for their support in regards to licenses.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.addma.2021.102193](https://doi.org/10.1016/j.addma.2021.102193).

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